

# Thread pools, *or how to share a bounded crew.*

*From `threading.Thread` to `concurrent.futures` — when a pool helps, and when it just hides the problem.*

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| TOPIC                  | LANGUAGE    | RUNTIME     |
|------------------------|-------------|-------------|
| Thread pools & futures | Python 3.12 | 75 min read |

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CH. 7 · INTRODUCTION

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A thread pool is a *bounded crew* of workers that take jobs from a shared queue. You hand it a function and its arguments; it hands back a **future** — a promise that the answer will be ready later. Pools solve two problems at once: they cap how many threads exist, and they remove the cost of starting a new one for every task.

This chapter assumes you have seen `threading.Thread` but have not yet reached Ch. 8 · `asyncio`. We stay deliberately in the `concurrent.futures` module, which is the right tool for the overwhelming majority of I/O-bound Python programs.<sup>1</sup>

**What you will learn**

|          |                                                                              |
|----------|------------------------------------------------------------------------------|
| Pool     | A fixed-size set of worker threads that consume a work queue.                |
| Future   | An object representing a computation that may not yet have completed.        |
| Executor | The object you submit work to; it owns the pool and the queue.               |
| GIL      | The Global Interpreter Lock; only one thread runs Python bytecode at a time. |

**A note on scope**

We cover *thread* pools specifically. Process pools ( `ProcessPoolExecutor` ) are touched on in §7.4 only to contrast sizing rules. *Asynce* pools are deferred to chapter 8.

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<sup>1</sup>For CPU-bound work, substitute `ProcessPoolExecutor` ; the API is identical, the cost model is not.

## 7.1 · THREADS &amp; THE GIL

## Why a pool, and why bounded.

Spawning a thread in Python is cheap but not free. Each thread carries an OS-level stack (8 MB on Linux by default), a bookkeeping structure in the interpreter, and contention for the **GIL**. Unbounded spawning is the single most common cause of a Python server going sideways under load.

### 7.1.1 Three reasons to pool

1. **Bound memory.** A fixed worker count caps stack usage to a predictable multiple of the stack size.
2. **Amortize startup.** Thread creation is 100 µs; reusing a worker for a 1 ms task matters.
3. **Backpressure for free.** When the queue fills, submitters block — the pool refuses to paper over a too-slow consumer.

### 7.1.2 When threads do *not* help

- Pure CPU work in pure Python — the GIL serializes it.
  - Use `ProcessPoolExecutor` instead.
  - Or drop into C via NumPy / `numba`.
- Work that already releases the GIL (e.g. a `requests` call) benefits regardless.
- Work that calls into a C extension holding a lock of its own.

*A thread pool is a queue in a trench coat. Everything interesting is in how the queue behaves when it is full, and in what the workers do when the queue is empty.*

### 7.1.3 Rules of thumb

- ☒ Is the workload I/O-bound?
- ☒ Am I willing to cap memory with a worker count?
- ☐ Can I tolerate out-of-order completion?

**The GIL in 3.13.** PEP 703 introduces a no-GIL build. Until it is the default, reason as if the GIL is there.

**Stack size.** Tunable via `threading.stack_size()` — rarely worth doing.

## 7.2 · WHAT A POOL ACTUALLY IS

(This section is listed in the chapter TOC but its content lives in the companion reference card; see Appendix A.)

## 7.3 · SUBMITTING &amp; COLLECTING WORK

## The minimal executor.

The standard library ships `concurrent.futures.ThreadPoolExecutor`. It is a context manager; entering it spins the workers, exiting it joins them.

fetch\_all.py

Python 3.12

```
from concurrent.futures import ThreadPoolExecutor, as_completed
import requests

urls = ["https://example.com/", "https://python.org/"]

def fetch(url: str) -> int:
    return requests.get(url, timeout=5).status_code
```

**Context manager.** Exiting the `with` block calls `shutdown(wait=True)`.

```
with ThreadPoolExecutor(max_workers=8) as ex:
    futures = {ex.submit(fetch, u): u for u in urls}
    for f in as_completed(futures):
        print(futures[f], f.result())
```

**NOTE**

Use `as_completed` when order does not matter — results arrive in finish order, not submit order. For submit order, iterate `ex.map` instead.

**TIP**

Wrap `f.result()` in `try/except`. Exceptions raised inside a worker are deferred until you ask for the result.

**WARNING**

Never `submit()` from inside a worker of the same pool — you can deadlock. Use a separate pool, or just call the function directly.

**DANGER**

Do not share an unsynchronised mutable (a `list`, `dict`) across workers. The GIL protects bytecode, not your invariants.

## 7.4 · SIZING THE POOL

## How many workers?



*Fig. 1* Submitters push callables into a FIFO work queue; a fixed set of workers pull from it.

A reasonable default for I/O-bound work is given by Little's law:

$$N = \lambda \cdot W \quad (1)$$

where  $N$  is the pool size,  $\lambda$  the arrival rate of requests, and  $W$  the average time a worker spends per request (mostly blocked on I/O).

**Little's law.** Kleinrock 1975; applies to any stable queueing system.

| Workload       | Good default | Ceiling | Why                |
|----------------|--------------|---------|--------------------|
| Local disk I/O | 4 – 8        | 32      | Kernel queue depth |
| HTTP calls     | 8 – 32       | 256     | Remote capacity    |
| DNS lookups    | 16           | 64      | Resolver cache     |

| Workload        | Good default | Ceiling | Why |
|-----------------|--------------|---------|-----|
| Pure Python CPU | 1            | 1       | GIL |

7.5 · EXERCISES

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## Work these before 7.6.

Model solutions are in Appendix C, pp. 342–346.

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# 01

SUBMIT / RESULT

**Warm-up.** Using `ThreadPoolExecutor`, compute the length of ten URLs in parallel and print them in *submission* order, not completion order.

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# 02

LITTLE'S LAW

**Sizing.** A service receives 40 requests/second; each spends 0.6 s blocked on a downstream API. Compute the pool size from equation Eq. 1. Justify rounding up.

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# 03

DEADLOCK

**Trap.** Construct a program where a worker in pool A submits to the same pool and waits on the result. Show the deadlock; fix it with two pools.

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*Concurrency is a way of organising code. Parallelism is a property of how it runs.*

— *Rob Pike · Concurrency is not parallelism (2012)*

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7.6 · FURTHER READING

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- **Kleinrock, L.** *Queueing Systems, Vol. 1*. Wiley, 1975.
- **Beazley, D.** *Python Concurrency From the Ground Up*. PyCon US, 2015.
- **CPython source** — `Lib/concurrent/futures/thread.py`.

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**See also.** Ch. 8 · `asyncio`;  
Ch. 9 · process pools.